6D_2 is the Case where the two curled dimensions form a sphere.

Notation and Setup:

Starting Metric for this Case: $g_{00}=\frac{5\,\dot{a}[t]^2}{-5\,k+\Lambda\,a[t]^2}$, $g_{11}=-\frac{a[t]^2}{\mathcal{D}_4[r]}$, $g_{22}=-r^2\,a[t]^2$, $g_{33}=-r^2\,a[t]^2\,\text{Sin}[\theta]^2$, $g_{44}=-a[t]^2\,p_4^2\,\mathcal{D}_4[r]$, $g_{55}=-a[t]^2\,\text{Sin}[\psi_4]^2\,p_4^2\,\mathcal{D}_4[r]$, a(t) not constant

Corresponding Line Element

$$ds^2 = \frac{5\,\dot{a}[t]^2}{-5\,k+\Lambda\,a[t]^2}\,dt^2 - \frac{a[t]^2}{\mathcal{D}_4[r]}\,dr^2 - r^2\,a[t]^2\,d\theta^2 - r^2\,a[t]^2\,\text{Sin}[\theta]^2\,d\phi^2 - a[t]^2\,p_4^2\,\mathcal{D}_4[r]\,d\psi_4^2 - a[t]^2\,\text{Sin}[\psi_4]^2\,p_4^2\,\mathcal{D}_4[r]\,d\psi_5^2$$

Output Display Simplification - Parameters are sometimes suppressed
Prime (if any) means derivative wrt r.
Dot (if any) means derivative wrt t.
Examples: **A = A(t,r)**, **$A'=\partial_r A(t,r)$** and **$\dot{A}=\partial_t A(t,r)$**

Lambda Vacuum Equations: The equations are $R_{\mu\nu} = \Lambda\, g_{\mu\nu}$. They are expressed as $\frac{R_{\mu\nu}}{g_{\mu\nu}} = \Lambda$ when $g_{\mu\nu} \neq 0$, and $R_{\mu\nu} = 0$ when $g_{\mu\nu} = 0$. The $g_{\mu\nu}$ matrix used is a diagonal matrix. Since off-diagonal $R_{\mu\nu}$ equations rarely appear, almost all the equations are expressed by dividing out $g_{\mu\nu}$.

GRW Equations: The exact solution to the Λ vacuum field equations that has (1) formulas for B, C, and $\mathcal{D}_4$, $\mathcal{D}_5$, ..., (2) a(t) as a free parameter, and (3) parameter $\mathcal{A}$ such that $\mathcal{A}(t) = c^2$ is the Friedmann Continuity Equation (which generates the a(t) formula).

a: GRW Equations

As always, we seek assumptions that can simplify the ODEs.

A pattern has been established in 4D, 5D,
and 6D_1 for initial assumptions for parameters A and C:

- $A(t,r) = \mathcal{A}(t) = \frac{5\,\dot{a}[t]^2}{-5\,k+\Lambda\,a[t]^2}$, where curvature k is a free parameter
- C(r) = 1, which is always the case in GRW equations

As for the pattern for B, the previous formulas for B have been:

4D: $B(r) = \frac{1}{1-k\,r^2}$

5D: $B(r) = \frac{1}{1 - k r^2 + \frac{\chi}{r}} = \frac{1}{\mathcal{D}_4[r]}$

6D_1: $B(r) = \frac{1}{1 - k r^2 + \frac{\chi}{r} + \frac{\Xi}{r^2}} = \frac{1}{\mathcal{D}_5[r]}$

(although k and Ξ were forced to be zero)

In 6D_2, there is only one curled object, a 2-sphere (S^2). Consequently, there is only a single $\mathcal{D}_\mu$ parameter, namely $\mathcal{D}_4$.

This suggests 3 possibilities for the $B(r)$ assumption:

(1) $B(r) = \frac{1}{\mathcal{D}_4[r]}$

(2) $B(r) = \frac{1}{\mathcal{D}_4[r]^2}$

(3) $B(r) = \frac{1}{\sqrt{\mathcal{D}_4[r]}}$

Options (2) and (3) arise because the prior patterns have been based on circles (S^1) and now the curled object is a sphere. This suggests that either $B = \frac{1}{\mathcal{D}_4{}^2}$ (option 2) or $\mathcal{D}_4 = \frac{1}{B^2}$ (option 3) may be a good choice.

However, options (2) and (3) can be ruled out: each forces $k = 0$ and its integration constant to zero, giving $\mathcal{D}_4 = B = 1$, and then R_{44} cannot be satisfied because of the curvature of the curled 2-sphere.

So, option (1) is the one we pursue.

In summary, our assumptions are:

- $A(t,r) = \mathcal{A}(t) = \frac{5\,\dot{a}[t]^2}{-5k + \Lambda\, a[t]^2}$, where k is a free parameter
- $B(r) = \frac{1}{\mathcal{D}_4[r]}$
- $C(r) = 1$,
- $\mathcal{D}_4(r)$ is a free parameter so we can solve for its formula.

The initial Lambda field equations are:

R_{00}: $\Lambda = \Lambda$

R_{11}: $-\frac{4k}{a^2} + \Lambda - \frac{\mathcal{D}_4'}{a^2 r} - \frac{\mathcal{D}_4''}{a^2} = \Lambda$

R_{22}: $-\frac{4k}{a^2} + \frac{1}{a^2 r^2} + \Lambda - \frac{\mathcal{D}_4}{a^2 r^2} - \frac{3\,\mathcal{D}_4'}{2\,a^2 r} = \Lambda$

$$R_{33}: \quad -\frac{4\,k}{a^2}+\frac{1}{a^2\,r^2}+\Lambda-\frac{\mathcal{D}_4}{a^2\,r^2}-\frac{3\,\mathcal{D}_4'}{2\,a^2\,r}=\Lambda$$

$$R_{44}: \quad -\frac{4\,k}{a^2}+\Lambda+\frac{1}{a^2\,p_4^2\,\mathcal{D}_4}-\frac{\mathcal{D}_4'}{a^2\,r}-\frac{(\mathcal{D}_4')^2}{4\,a^2\,\mathcal{D}_4}-\frac{\mathcal{D}_4''}{2\,a^2}=\Lambda$$

$$R_{55}: \quad -\frac{4\,k}{a^2}+\Lambda+\frac{1}{a^2\,p_4^2\,\mathcal{D}_4}-\frac{\mathcal{D}_4'}{a^2\,r}-\frac{(\mathcal{D}_4')^2}{4\,a^2\,\mathcal{D}_4}-\frac{\mathcal{D}_4''}{2\,a^2}=\Lambda$$

R_{33} and R_{55} are redundant, the same as R_{22} and R_{44}, respectively.

R_{00} and R_{01} (not listed) are identities due to the assumed $\mathcal{A}[t]$ formula.

We can solve R_{22} for $\mathcal{D}_4$:

$$\mathcal{D}_4(r)=1-k\,r^2+\frac{\chi}{r^{2/3}} \quad \text{where } \chi \text{ is a constant of integration.}$$

The remaining 2 equations become

$$R_{11}: \quad \Lambda-\frac{4\,\chi}{9\,a^2\,r^{8/3}}=\Lambda$$

$$R_{44}: \quad -\frac{k\,r^{2/3}}{a^2\left(r^{2/3}-k\,r^{8/3}+\chi\right)}+\frac{r^{2/3}\,\Lambda}{r^{2/3}-k\,r^{8/3}+\chi}-\frac{k\,r^{8/3}\,\Lambda}{r^{2/3}-k\,r^{8/3}+\chi}-\frac{16\,k\,\chi}{9\,a^2\left(r^{2/3}-k\,r^{8/3}+\chi\right)}+$$
$$\frac{\chi}{9\,a^2\,r^2\left(r^{2/3}-k\,r^{8/3}+\chi\right)}+\frac{\Lambda\,\chi}{r^{2/3}-k\,r^{8/3}+\chi}+\frac{r^{2/3}}{a^2\left(r^{2/3}-k\,r^{8/3}+\chi\right)p_4^2}=\Lambda$$

$R_{11} \Rightarrow \chi = 0$

$$\Rightarrow \quad R_{44}: \ \Lambda+\frac{-1+k\,p_4^2}{\left(-1+k\,r^2\right)a[t]^2\,p_4^2}=\Lambda$$

$$\Rightarrow \quad k=\frac{1}{p_4^2}>0$$

All of the equations are satisfied. The 6D_2 GRW solution is

$$\mathcal{A}(t)=\frac{5\,\dot{a}[t]^2}{-5\,k+\Lambda\,a[t]^2}=c^2 \quad \text{where } k=\frac{1}{{p_4}^2}>0,$$

$$\mathcal{B}(r)=\frac{1}{1-k\,r^2},$$

$$\mathcal{C}(r)=1,$$

$$\mathcal{D}_4(r)=1-k\,r^2=\frac{1}{\mathcal{B}(r)},$$

and line element

$$ds^2=c^2\,dt^2-\frac{a[t]^2}{1-k\,r^2}\,dr^2-r^2\,a[t]^2\,d\theta^2-r^2\,a[t]^2\,\text{Sin}[\theta]^2\,d\phi^2-\left(1-k\,r^2\right)a[t]^2\,p_4^2\,d\psi_4^2$$
$$-\left(1-k\,r^2\right)a[t]^2\,\text{Sin}[\psi_4]^2\,p_4^2\,d\psi_5^2$$

Geometrically, this is the familiar closed (k > 0) FLRW universe in five spatial dimensions. Substituting r = p_4 sin ψ ($0 \le \psi \le \pi/2$), the spatial part of the line element becomes

$$-a(t)^2\, p_4{}^2\, [\, d\psi^2 + \sin^2\psi\, (d\theta^2 + \sin^2\theta\, d\phi^2) + \cos^2\psi\, (d\psi_4{}^2 + \sin^2\psi_4\, d\psi_5{}^2)\,],$$

which is exactly a round 5-sphere (S^5) of radius p_4. This is why $k = 1/p_4{}^2$.

Because B(r) depends on $k = \frac{1}{p_4{}^2}$, the size of the base 3-space, which should be either very large or infinite, is limited by p_4.

This solution cannot represent our universe.

Remark: Why a curved compact space ties k to its radius

Express the spatial part of the metric as

$$ds^2 = -a(t)^2\, h,$$

where h contains everything that doesn't depend on t (B, C, D_4, ...).

After $\mathcal{A}(t)$ is substituted, we see above that every spatial field equation R_{ii} contains the same term, $-\frac{4\,k}{a^2}$. Thus

$$\frac{R_{ii}\,(h)}{h_{ii}} = 4k$$

for every spatial direction i = 1-5 (and the off-diagonal $R_{ij} = 0$).

$R_{ij}(h)$ is the Ricci tensor of the 5-dimensional spatial metric h alone.

Definition. For a diagonal metric, the ratio $\frac{R_{ii}}{h_{ii}}$ is called the **Ricci curvature in direction i**.

Geometrically, Ricci curvature in direction i is the sum of the curvatures of all the 2-dimensional planes that contain direction i.

In 5 dimensions there are 4 such planes, which is where the 4 comes from.

- Adding it up over all directions gives the familiar curvature scalar, $R = 5 \cdot 4k$.

Example. On a sphere of radius p, every 2-plane has curvature $\frac{1}{p^2}$,

so a 5-sphere has Ricci curvature $\frac{4}{p^2}$ in every direction.

Consequence.

- The field equations force the Ricci curvature to be the same number, 4k, everywhere and in every direction.
- One constant, k, controls the curvature of the whole space:
  - ✓ Ordinary 3-space
  - ✓ Curled 2-space (S²)

Why this matters for 6D_2.

- The curled S^2 of radius p_4 is curved, with curvature $\frac{1}{{p_4}^2}$.
- Its directions must still have Ricci curvature 4k, so k can't be independent of p_4.
- The whole 5-space is a 5-sphere of radius p_4, with Ricci curvature $\frac{4}{{p_4}^2}$, so $4k = \frac{4}{{p_4}^2}$ and $k = \frac{1}{{p_4}^2}$.
- Ordinary 3-space then shares that curvature, and the universe is closed with size about p_4.
- (A general theorem, Myers' theorem, says the same thing:
  - ✓ Positive Ricci curvature 4k in every direction forces a finite space, at most $\frac{\pi}{\sqrt{k}}$ across.)

Contrast with circles.

- A circle has no curvature, so a curled circle (5D, 6D_1, 7D_1) doesn't constrain k this way.
- That's why the circle cases allow k = 0 or k free, while the sphere cases (6D_2, 7D_2 Solution 1, 7D_3) force a closed universe tied to the sphere's radius.

b: Solve $\mathcal{R}(t) = c^2$ to find a(t)

The steps for finding a(t) by solving $\frac{5\,\dot{a}[t]^2}{-5\,k + \Lambda\,a[t]^2} = c^2$ were provided in Case 4D_a.

They result in a solution with 4 branches, only one of which includes k>0:

$$a(t) = \sqrt{5}\,\sqrt{\frac{1}{\Lambda}}\,\sqrt{|k|}\,\cosh\left(\frac{c\,\sqrt{\Lambda}\,t}{\sqrt{5}}\right)$$

c: Kaluza-Klein (KK) bubble-of-nothing

In Case 6D_2, the curled S^2 does pinch to a point at $r_0=p_4$, as in a KK bubble, but no smoothness condition is needed:

- In order for $\mathbb{D}_4(r) \geq 0$, we must have $r^2 \leq p_4^2$, or $0 \leq r \leq p_4$.
- The "pinch" candidate would be $r_0 = p_4$:
  - ✓ The domain of r is restricted to $0 \leq r < r0$
  - ✓ The radial line element for this solution is

$$ds^2 = \frac{1}{\mathbb{D}_4[r]}\, dr^2 + \mathbb{D}_4[r]\; p_4^2\; [d\psi_4^2 + \text{Sin}[\psi_4]^2\; d\psi_5^2]$$

  - ✓ As $r \to r0$
    - $B \to \infty$ and $\mathbb{D}_4 \to 0$
    - The whole (ψ_4, ψ_5) two-sphere shrinks from a sphere of radius p_4 smoothly down to sphere of radius zero, a single point
    - The 'bubble' closes up automatically without needing any constraints

There is no KK-bubble constraint on p_4. It remains a free modulus

d: Free-gauge GRW

The table below summarizes the GRW and FG-GRW solutions for 4D through 6D. The 6D_2 GRW equations were discovered in Section a. The FG-GRW equations are generated below the table.

		4D	5D	6D_1	6D_2
GRW	Con-straints	None	None	$\Lambda>0,\ k=0$	$k=\frac{1}{p_4^2}>0,\ \chi=0$
	KK-bubble Con-straints (optional)	n/a	$\chi=r_0\left(-1+k r_0^2\right)$ $r_0=\frac{\mp 1+\sqrt{1+3\,k\,p_4^2}}{3\,k\,p_4}$ $k\geq-\frac{1}{3\,p_4^2}$	$\chi=-\frac{p_5}{2}$	None
	$\mathcal{A}$	$\mathcal{A}(t)=c^2$	$\mathcal{A}(t)=c^2$	$\mathcal{A}(t)=c^2$	$\mathcal{A}(t)=c^2$
	B	$\frac{1}{1-k r^2}$	$\frac{1}{\mathcal{D}_4}$	$\frac{1}{\mathcal{D}_5}$	$\frac{1}{\mathcal{D}_4}$
	C	1	1	1	1
	$\mathcal{D}_4$		$1-k r^2+\frac{\chi}{r}$	1	$1-k r^2+\frac{\chi}{r^{2/3}}$
	$\mathcal{D}_5$			$1-k r^2+\frac{\chi}{r}$	
FG GRW	$\mathcal{A}$	$\frac{3\,\dot{a}^2}{-3\,k+a^2\,\Lambda}$	$\frac{4\,\dot{a}^2}{-4\,k+a^2\,\Lambda}$	$\frac{5\,\dot{a}^2}{-5\,k+a^2\,\Lambda}$	$\frac{5\,\dot{a}^2}{-5\,k+a^2\,\Lambda}$
	B	$\frac{(2\,C+r\,C')^2}{4\,C\,\left(1-k\,r^2\,C\right)}$	$\frac{(2\,C+r\,C')^2}{4\,C\mathcal{D}_4}$	$\frac{(2\,C+r\,C')^2}{4\,C\mathcal{D}_5}$	$\frac{(2\,C+r\,C')^2}{4\,C\mathcal{D}_4}$
	C	C (free)	C (free)	C (free)	C (free)
	$\mathcal{D}_4$		$1-k\,r^2\,C+\frac{\chi}{r\,\sqrt{C}}$	1	$1-k\,r^2\,C+\frac{\chi}{\sqrt[3]{r^2\,C}}$
	$\mathcal{D}_5$			$1-k r^2 C+\frac{\chi}{r\,\sqrt{C}}$	

Notes:

- $\mathcal{A}(t)=c^2$ is the Friedmann Continuity Equation
  - ✓ The solution to the equation is the $a(t)$ formula from Section b
- The two constraint rows display restrictions on parameter constants
  - ✓ Due to the field equations
  - ✓ Due to KK-bubble smoothness
- By contrast, domain restrictions on r (RCPC and its analogs) are not displayed as constraints
- FG-GRW has the same constraints (both rows) as GRW

The 6D_2 Free-gauge GRW results seen in the table were obtained by applying $r^2\to r^2 C$, the pattern established in 4D, to the GRW parameters generated in Section a. Specifically,

$\mathcal{A}$: Replace $\mathcal{A}(t)=c^2$ by the $\mathcal{A}(t)$ formula

$\mathbb{B}$: Replace $r^2 \to r^2\, \mathbb{C}$ and multiply result by $\frac{(2\,\mathbb{C} + r\,\mathbb{C}')^2}{4\,\mathbb{C}}$

$\mathbb{C}$: Replace 1 by free parameter $\mathbb{C}$

$\mathbb{D}_4$: Replace r^2 by $r^2\mathbb{C}$

The line element is

$$ds^2 = \frac{5\,\dot{a}[t]^2}{-5\,k + \Lambda\, a[t]^2}\, dt^2 - \frac{a[t]^2\,(2\,\mathbb{C}[r] + r\,\mathbb{C}'[r])^2}{4\,\mathbb{C}[r]\,\left(1 - k\,r^2\,\mathbb{C}[r]\right)}\, dr^2 - r^2\, a[t]^2\,\mathbb{C}[r]\, d\theta^2$$
$$- r^2\, a[t]^2\, \text{Sin}[\theta]^2\,\mathbb{C}[r]\, d\phi^2$$
$$- a[t]^2\, p_4^2\,\left(1 - k\,r^2\,\mathbb{C}[r]\right)\, d\psi_4^2$$
$$- a[t]^2\, \text{Sin}[\psi_4]^2\, p_4^2\,\left(1 - k\,r^2\,\mathbb{C}[r]\right)\, d\psi_5^2$$

To confirm these equations, we confirm that the field equations are satisfied:

R_{00}: $\Lambda = \Lambda$

R_{01}: $0 = 0$

R_{02}: $0 = 0$

R_{03}: $0 = 0$

R_{04}: $0 = 0$

R_{05}: $0 = 0$

R_{11}: $\Lambda = \Lambda$

R_{12}: $0 = 0$

R_{13}: $0 = 0$

R_{14}: $0 = 0$

R_{15}: $0 = 0$

R_{22}: $\Lambda = \Lambda$

R_{23}: $0 = 0$

R_{24}: $0 = 0$

R_{25}: $0 = 0$

R_{33}: $\Lambda = \Lambda$

R_{34}: $0 = 0$

R_{35}: $0 = 0$

R_{44}: $\Lambda = \Lambda$

R_{45}: $0 = 0$

R_{55}: $\Lambda = \Lambda$

e: The Vacuum Friedmann Equations

Plugging $\mathcal{A}$, B, C, and D_4 (but not a)
from the GRW solution into the field equations yields:

R_{00}: $\frac{5\,\ddot{a}}{a\,c^2} = \Lambda$

R_{11}: $\frac{4\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} + \frac{4}{a^2\,p_4^2} = \Lambda$

R_{22}: $\frac{4\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} + \frac{4}{a^2\,p_4^2} = \Lambda$

R_{22} is redundant, the same as R_{11}.

The Friedmann continuity and acceleration equations, respectively, are the
following pair of equations generated by algebraic manipulation of R_{00} and R_{11}:

$-\frac{1}{20}R_{00} + \frac{1}{4}R_{11}$: $\frac{\dot{a}^2}{a^2\,c^2} + \frac{1}{a^2\,p_4^2} = \frac{\Lambda}{5}$

$\frac{1}{6}R_{00} + \frac{1}{6}R_{11}$: $\frac{2\,\dot{a}^2}{3\,a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} + \frac{2}{3\,a^2\,p_4^2} = \frac{\Lambda}{3}$